

ADAPTIVE EXECUTION & ROUTING OPTIMIZATION

The right execution. For every workload.

Product and Technology Guide

A comprehensive guide to policy-driven routing, cost control, governed execution and continuous optimization for enterprise AI and data platforms.

PROFILE Understand the work → **ROUTE** Apply policy and budget → **EXECUTE** Use the eligible lane → **IMPROVE** Replay, review and learn

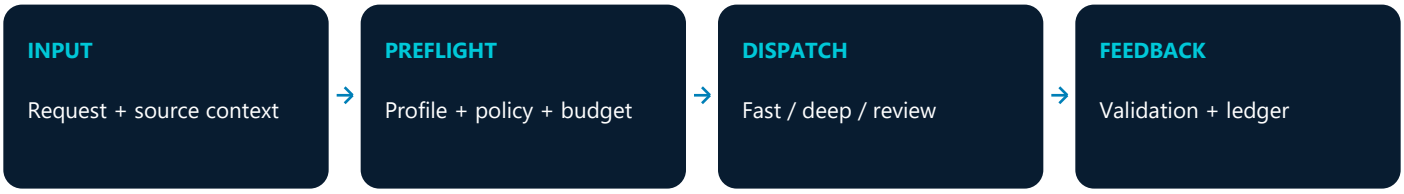
Microsoft Fabric | Azure | AI and data pipelines

PRODUCT REFERENCE EDITION / OCTOBER 2026

One decision plane. Multiple paths.

AERO separates execution decisions from the systems that perform the work.

AERO evaluates a workload before it consumes expensive compute or premium inference. It profiles complexity and risk, estimates execution cost, applies governance policy and selects Fast Path, Deep Path or Review Path. Every decision carries an explanation and a policy version.



How the platform fits together

LAYER	RESPONSIBILITY
Control plane	FastAPI accepts requests; Pydantic validates contracts; the profiler, policy catalog, router and BudgetGuard authorize execution.
Execution plane	Adapters dispatch work to SQL, Python, Fabric pipelines and notebooks, Azure Databricks, approved AI providers or review workflows.
Evidence plane	Validation, tracing and the execution ledger capture outcomes, cost, latency, provenance and reviewer feedback.
Optimization plane	Replay Lab compares policies; Policy Diff explains the change; Autopilot turns evidence into approval-controlled recommendations.

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Profile early. Route deliberately.

The profiler creates a structured view of the workload; policy determines what that workload is allowed to do.

EXECUTION LANE	WHEN IT FITS	TYPICAL EXECUTION
Fast Path	Routine, bounded, low-risk work with sufficient evidence and compatible inputs.	Warehouse SQL, lightweight Python, deterministic rules, cache reuse or a smaller approved model.
Deep Path	Large joins, schema drift, semantic difficulty, cross-source dependencies or high business impact.	Fabric Spark / PySpark, Azure Databricks, stronger AI models and deeper validation.
Review Path	Restricted data, conflicting evidence, low confidence, failed checks or required human approval.	Approval queue, steward review, quarantine or a governed policy exception.

What the preflight profile contains

DIMENSION	EXAMPLES
Workload and source	Workload type, application, team, source system and input reference.
Shape and complexity	Record count, schema compatibility, drift, partition size and dependency signals.
Risk and evidence	Sensitivity, confidence, ambiguity signals, business impact and approved guidance.
Service and cost	SLA or freshness target, estimated work units, eligible engines and predicted execution cost.
Validation plan	Contract, schema, row-count, semantic, sensitivity and human-approval checks.

Review has precedence
A difficult workload can also be sensitive or uncertain. AERO retains all escalation evidence, while review requirements prevent automatic dispatch.

Example: a schema-compatible mapping below its configured size threshold is eligible for Fast Path. A schema change escalates it to Deep Path. Restricted sensitivity or conflicting source evidence sends it to Review Path even when a compute engine could process it.

Policy authorizes. BudgetGuard controls.

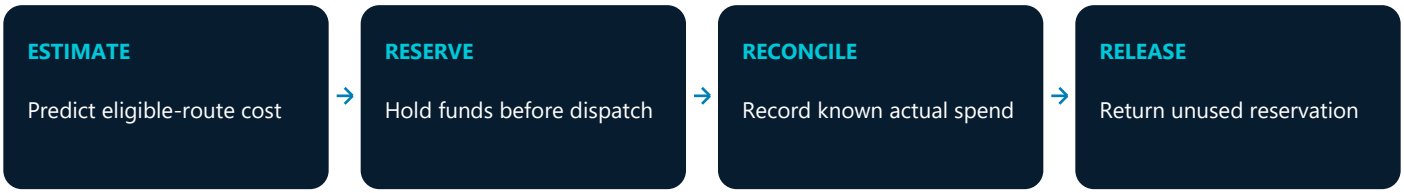
Versioned rules make execution behavior inspectable, repeatable and accountable.

Routing policies describe default lanes, eligible engines, fast-path size and sensitivity limits, deep-path signals, review triggers and reservation ceilings. AI-assisted classifications remain recommendations; deterministic policy and governance checks authorize the route.

Policy decisions in practice

CONDITION	AERO BEHAVIOR
Routine + eligible	Use the configured efficient lane and required validation plan.
Complex or high-impact	Escalate to the deeper execution engine with an explanation.
Uncertain or governed	Create a review case with evidence, policy version and route reason.
Reservation above a ceiling	Block dispatch and record the attempted route and estimated cost.
Unsupported engine or invalid input	Reject dispatch or quarantine the input with a traceable reason.

The budget lifecycle



BudgetGuard compares the request estimate with policy limits, workload-specific ceilings and available budget. A successful reservation protects capacity for that execution. Reconciliation records known spend, including costs incurred by failed attempts; unused reservations are released.

Cost has a model and a provenance
Track estimated, reserved and actual or reconciled amounts separately. Rate versions, measured usage and the workload family explain the estimate. Fabric capacity consumption is translated into an allocation model for planning; workload dollar estimates are distinct from a provider invoice. [3]

Change discipline: version the policy, replay it against representative workloads, inspect the diff, obtain approval, promote the selected version and retain its rollback reference.

Test change before it reaches production.

Replay Lab, Policy Diff Simulator and Autopilot With Guardrails make optimization evidence-driven.

AERO Replay Lab

Replay Lab runs the same historical, synthetic or projected workload set through current and candidate policies. It shows lane distribution, estimated spend, savings against a deep-only eligible baseline, review burden, budget blocks and validation-risk notes. Each replay preserves its workload set, policy versions and assumptions.

ILLUSTRATIVE: 1,000 WORKLOADS	DEEP BASELINE	CURRENT	CANDIDATE
Fast executions	0	650	700
Deep executions	880	230	165
Review cases	100	100	110
Blocked workloads	20	20	25
Estimated execution cost	$6,000	$2,600	$2,250
Savings vs. deep baseline	-	56.7%	62.5%

Illustrative simulation only. The four outcome groups total 1,000 in each policy. Cost figures are sample projections; review labor, quality risk and operational overhead are evaluated separately.

Policy Diff Simulator

The diff identifies changed routes by workload ID, added or reduced spend, additional or reduced review cases, newly blocked work and validation-risk changes. In the example above, the candidate projects $350 less execution spend, 10 additional review cases and 5 additional blocks than the current policy.

Autopilot With Guardrails

Autopilot recommends cheaper eligible routes, stronger review triggers, budget adjustments, additional validation or team-specific policy variants. Each recommendation includes evidence, expected impact, risk and rollback notes. Recommendations start **pending** and can be **approved** or **rejected**; approval and policy deployment are explicit decisions.

A lower projection is one input to a decision
Approve changes using cost, validation evidence, service targets and review capacity together. Replay identifies the tradeoff; people determine whether the change is acceptable.

A practical, extensible technology stack.

AERO combines a Python control plane with adapter-based execution and environment-appropriate storage.

TECHNOLOGY	ROLE IN AERO	WHY IT FITS
Python	Profiling, routing, adapters, replay and reporting.	Shared implementation language across data engineering and AI workloads.
FastAPI [5]	HTTP boundary for workload submission and service integration.	Typed request handling and API documentation at the control-plane boundary.
Pydantic	Workload, decision, result and recommendation contracts.	Strict structured validation before data can influence routing or execution.
YAML	Versioned routing policy catalog.	Readable policy rules, reviewable diffs and reproducible policy releases.
SQLite / managed SQL	Execution ledger and operational state.	SQLite supports local operation; Azure SQL Database or PostgreSQL supports hosted storage.
Streamlit [6]	Operational dashboard and policy exploration.	Python-based views for spend, lane mix, review queues and replay results.
uv, pytest and Ruff	Dependency management, behavior tests and code quality.	Repeatable environments and checks for routing, budgets, review and replay.
OpenTelemetry [7]	Traces and correlated operational signals.	Connects profiling, dispatch and validation to environment-specific monitoring.

Execution and platform integrations

Data processing: Fabric Warehouse SQL, Dataflows Gen2, Fabric Spark notebooks, Azure Databricks, SQL engines and local Python adapters.

AI execution: Azure OpenAI, OpenAI or another approved provider behind an adapter. Model aliases, request ceilings, structured output validation and governance remain part of AERO policy.

Cloud services: Azure Container Apps or App Service for hosting; OneLake, ADLS Gen2 or Blob Storage for data; Microsoft Entra ID, Key Vault and Microsoft Purview for environment controls; Azure Monitor and Application Insights for operations.

Built around Fabric and Azure workflows.

The decision service complements pipelines, lakehouses, warehouses and approved execution engines.

FABRIC INPUT
Pipeline or notebook context

→

AERO GATE
Profile, authorize, reserve

→

EXECUTION
SQL / Spark / AI / review

→

EVIDENCE
Validate, reconcile, record

PLATFORM COMPONENT	HOW AERO USES IT
OneLake and Lakehouse [1]	Workload references, source metadata and Delta-based data engineering connect the decision plane to shared analytical data.
Fabric Data Factory	Pipelines submit preflight context and branch on the approved lane. Dataflows Gen2 provide a transformation option for eligible workloads.
Warehouse SQL and Spark [2]	Warehouse SQL handles relational execution; Spark handles lakehouse writes and complex transforms. The lakehouse SQL analytics endpoint is a read-only query surface.
Azure Container Apps [4]	Hosts the API and dashboard containers. App Service provides another hosted web-service option.
Managed storage and secrets	Azure SQL Database or PostgreSQL persists operational state; Key Vault supplies secrets; managed identities limit credential exposure. [8]
Identity and observability	Microsoft Entra ID provides identity; Azure Monitor and Application Insights capture service telemetry. Fabric capacity monitoring supplies platform utilization context. [3]

Keep responsibilities clear
Fabric orchestrates data movement and execution. AERO authorizes the lane and captures evidence. Source permissions, OneLake controls and execution-engine identities remain enforced by the underlying platform.

Deployment profiles: local reference installation for policy testing; an internal hosted service for shared operation; or an enterprise environment with private connectivity, managed identity, durable state, approved adapters and environment-specific policy catalogs.

Governed execution includes the outcome.

Validation, review feedback and quality controls close the loop after a routing decision.

CAPABILITY	WHAT IT DOES
Validation	Checks structured output, source and result schemas, row counts, semantics and policy-specific acceptance conditions. Failures change status or create review work.
Approved evidence	Retrieves approved runbooks, mapping standards and policy guidance. Deterministic calibration applies hard constraints after AI-assisted recommendations.
Review workflow	Stores workload ID, route reason, evidence, policy version and reviewer disposition: accepted, edited or rejected.
Learning loop	Groups reviewer reasons such as wrong lane, weak evidence, validation issue or policy gap into a policy and evaluation backlog.
Quarantine and triage	Separates invalid inputs and failed partitions; records source and failure reason; generates a report explaining likely cause and next action.
Last-known-good release	Retains validated run or policy metadata so teams can compare a release with an accepted reference and restore a known version.
Mapping Workbench	Examines source-to-target mapping decisions, simulates transformations and preserves mapping trace and audit notes.

An explainable review case

A mapping job contains a changed schema and restricted fields. The profile retains both facts; policy chooses Review Path. The reviewer receives the source context, required checks and route explanation. An edited disposition identifies the approved mapping change, and the feedback becomes an evaluation case before the policy is tuned.

Separate policy approval from data access
Policy owners govern route rules; reviewers resolve individual cases; platform administrators govern identities and access. Sensitive payloads stay behind authorized data references, while the ledger records the evidence needed to explain a decision.

Practical uses. Measurable operation.

AERO applies one decision framework across data processing, AI execution and operational exceptions.

USE CASE	EXECUTION PATTERN	VALUE TO MEASURE
Incremental data loads	Routine partitions use lightweight transforms; large or changed partitions escalate.	Compute per validated partition, freshness and selective recompute.
AI classification and enrichment	Simple classifications use an efficient model; nuanced cases use deeper inference or review.	Cost per acceptable result, escalation rate and output quality.
Data quality incidents	Approved guidance supports routine triage; ambiguous or sensitive failures enter review.	Resolution latency, recurrence, quarantine volume and review outcomes.
Source-to-target mappings	Simulate mappings, validate schema compatibility and retain a transformation trace.	Schema stability, mapping exceptions and trace completeness.
Governed reporting	High-impact summaries receive deeper validation; restricted or approval-required work is held.	Validation coverage, approval turnaround and decision provenance.
Policy and capacity tuning	Replay a representative cohort, inspect the diff and approve an evidence-backed change.	Projected spend, capacity demand, new blocks and review burden.

Operate from the dashboard and ledger

The dashboard shows estimated and reconciled spend, lane mix, cost drivers, review backlog, validation failures, budget blocks and policy comparisons. A ledger record connects workload and run IDs to team, source, engine, policy version, route reason, attempts, latency, validation and review status.

Metrics that support decisions
Track savings against the same eligible workload cohort, median and slowest latency, validation pass rate, failed and blocked rate, review turnaround and cost per acceptable completed result. Keep estimates, measured outcomes and projected workload sets identifiable.

Troubleshooting: start with the run ID. Inspect the route reason and policy version, confirm reservation outcome, check adapter attempts and validation details, then follow source or review evidence to the next action.

Integrate through a workload contract.

A common request structure carries the context needed for profiling, policy, cost control and auditing.

```
{
  "workload_id": "orders-daily-2026-10-12",
  "app_name": "orders_pipeline",
  "team_id": "analytics",
  "workload_type": "data_transform",
  "input_ref": "data/raw/orders.jsonl",
  "route_policy": "aero_routed",
  "max_cost_usd": 2.0,
  "metadata": {
    "source_system": "orders",
    "record_count": 12000,
    "schema_change_detected": false,
    "sensitivity": "internal",
    "confidence": 0.95,
    "sla_minutes": 30
  }
}
```

Example request. Policies and engine aliases are deployment-specific. The response explains the lane, policy version, reservation and validation requirements; the execution result adds status, cost, latency and review disposition.

Start with a representative workload set

1. Define sources, workload families, ownership and sensitivity.
2. Configure approved engines, validation checks and budget ceilings.
3. Establish a deep-only eligible baseline and replay candidate policies.
4. Review route changes, cost assumptions and review capacity.
5. Approve a policy version, connect the execution adapter and monitor outcomes.

Technical references

[1] Microsoft Fabric platform

[2] Lakehouse SQL analytics endpoint

[3] Fabric Capacity Metrics app

[4] Azure Container Apps

[5] FastAPI

[6] Streamlit

[7] OpenTelemetry

[8] Container Apps security and managed identity